Computers & Graphics (2026)

Contents lists available at ScienceDirect

Computers & Graphics

journal homepage: www.elsevier.com/locate/cag

Timestep-Conditioned Adapter Scaling for Multi-view Diffusion Texture Generation

ARTICLE INFO

Keywords: multi-view diffusion, 3D texturing, adapter scaling, timestep control, shape-texture trade-off

ABSTRACT

Multi-view diffusion pipelines for 3D asset texturing commonly inject geometric conditions through adapter modules, but adapter scale is not a universal guidance strength. We study Timestep-Conditioned Adapter Scaling (TCAS) as a training-free, three-stage inference intervention: a low–high–low schedule is selected on a probe set and then evaluated under frozen protocols. On a strict 276-object main-adapter holdout, a clean-v2 four-condition run and a separately labeled stage-placement follow-up show that stage placement changes the output, but no schedule dominates all reported targets. In the follow-up, LHL improves on selected targets relative to fixed mean and HLL, whereas LLH is better than LHL on all six non-SSIM evaluation metrics and on the saved-artifact Full-SSIM branch. A second-backbone audit likewise leaves the pre-specified CAI selection undefined. A 12-object stratified baking case study verifies the end-to-end GLB/unseen-view path but does not establish a population-level 3D advantage. We therefore present TCAS as a bounded, adapter-dependent mechanism study rather than a universal schedule-selection rule.

1. Introduction

Recent advances in single-image 3D generation have substantially improved the recovery of geometry, multi-view consistency, and sparse-view reconstruction [1, 2, 3, 4, 5, 6, 7, 8]. As geometric generation models become increasingly mature, generating high-quality textures on an existing 3D shape has become a key factor affecting the realism and usability of 3D content. For an object whose geometry has already been determined, the final visual quality depends not only on whether the shape is complete, but also on whether the texture is faithful to the reference image, aligned with the 3D surface, and rich in local details.

Existing multi-view texture generation methods usually generate RGB images from multiple target views using 2D image diffusion models, and then map the generated images back to the 3D surface through projection or texture baking [9, 10, 11, 12]. This route benefits from the generative capability of image diffusion models, but it still faces challenges in reference appearance alignment, geometry-texture consistency, and local texture detail preservation. Geometry-controlled multi-view diffusion methods represented by MVPainter introduce geometric conditions such as normals and depth maps to improve the alignment between generated textures and the underlying 3D surface, demonstrating the importance of explicit geometric control for 3D texture generation [13].

However, introducing geometric conditions is not only a question of whether a control signal should be used, but also a question of how strongly it should be applied. In adapter-augmented diffusion pipelines, geometric conditions are usually injected into the model through an adapter or control branch, and their influence is regulated by a scaling factor. In practice, this factor is often set empirically: a small scale may lead to insufficient geometric constraints, whereas a larger scale is usually expected to improve structural consistency.

In multi-view texture generation, however, stronger adapter scaling is not necessarily better. Texture generation requires not only contours and structures to be consistent with geometry, but also surface color variations, material details, and high-frequency patterns to be preserved. Overly strong geometric control may force the model to follow normal, depth, or edge conditions too aggressively, thereby suppressing the intrinsic texture synthesis capability of the base diffusion model. In this case, some structural metrics may still improve, while the visual result suffers from texture flattening, reduced color variation, and detail loss.

Therefore, adapter scaling in geometry-controlled multi-view diffusion deserves further analysis. Compared with prior work that mainly studies how to build texture generation pipelines, how to introduce geometric conditions, and how to improve overall generation quality, adapter strength is a fine-grained inference-time control problem. It directly affects the balance between shape consistency and texture detail, and is an easily overlooked factor in multi-view texture generation quality.

This problem is related to, but not solved by, existing guidance scheduling strategies. CFG scheduling changes the extrapolation between conditional and unconditional predictions, while a geometry adapter injects residuals into intermediate representations. Generic ControlNet or adapter scale scheduling similarly treats condition strength mainly as a condition-adherence knob. In texture generation, however, the key question is not only how strongly to follow the geometry, but also when geometric residuals should dominate and when they should recede so that the base model can synthesize color, material, and high-frequency texture. Simple monotonic schedules such as linear or cosine decay do not explicitly address this non-monotonic requirement. Even the recent finding that classifier-free guidance is most useful only in a limited noise interval [14] does not answer this question: it identifies where conditioning helps for CFG, but not which stages of a geometry-conditioned texture pipeline require strong geometric residuals, where the late-stage risk is texture suppression rather than diversity loss.

The goal of this paper is not to propose a new full texture generation system, but to analyze and improve inference-time adapter scaling within a fixed geometry-controlled multi-view diffusion pipeline.

The main contributions of this paper are as follows:

(1) We systematically diagnose the shape-texture trade-off caused by adapter scaling in multi-view diffusion texture generation, showing that higher scales can improve structural metrics while simultaneously causing adapter-dependent texture degradation.

(2) We formulate Timestep-Conditioned Adapter Scaling (TCAS), a low–high–low inference schedule that concentrates strong geometric correction in the middle denoising stage while using conservative scales otherwise, and test its stage-utility hypothesis with controlled placement comparisons.

(3) We provide a strict-276 main-adapter audit with explicit pre-save versus saved-artifact metric provenance, a fixed-GT Full-SSIM serialization decomposition, and a 17/16/17-step budget audit. The resulting evidence identifies stage sensitivity but also a clear LLH counterexample to a universal LHL optimum.

(4) We assess practical 3D relevance with a traceable 12-object, four-condition GLB baking and unseen-view case study, and audit transfer to a second backbone while keeping cross-backbone scores and undefined CAI outcomes separate.

2. Related Work

2.1. 3D Texture Generation and Multi-view Diffusion

Single-image 3D generation is often decomposed into geometry generation and texture generation. In recent years, single-image-to-3D, multi-view diffusion, and sparse-view reconstruction methods have made notable progress in structural stability, shape completeness, and cross-category generalization [1, 2, 3, 4, 5, 6, 7, 8]. Nevertheless, high-quality texture generation remains a key bottleneck for the realism and usability of 3D content, as reflected by recent surveys of neural 3D mesh texturing [15]. Existing texture generation methods usually rely on 2D diffusion models to synthesize RGB images from multiple views, and then project or bake these images onto the 3D surface [9, 10, 11, 12]. While this strategy inherits the strong generative capability of image diffusion models, it still suffers from reference appearance misalignment, geometry-texture inconsistency, and insufficient local texture detail. Beyond these diffusion pipelines, the graphics community continues to advance surface appearance synthesis more broadly, from exemplar-based texture synthesis [16] to diffusion-based procedural material generation [17].

To address these issues, methods such as Paint3D, MVPaint, Hunyuan3D 2.0, and MVPainter formulate 3D texture generation as a geometry-controlled multi-view diffusion task, generating multi-view images under geometric control and baking them into UV textures [18, 19, 20, 13]. MV-Adapter further shows that adapter-style multi-view control can enhance cross-view consistency without fully retraining the base model [21], and recent texturing systems strengthen consistency during texture synthesis itself, for example through synchronized multi-view diffusion [22] or consistent-and-seamless text-guided multi-view generation [23]. By injecting geometric information such as normals and depth maps into the diffusion model as conditional signals, the generated results better follow the 3D surface structure and remain consistent across views. Unlike prior methods that focus mainly on multi-view image generation or general 3D reconstruction, these works emphasize precise alignment between texture and geometry as well as local detail recovery, significantly improving 3D texture generation quality. However, they mainly answer how to introduce geometric conditions and how to construct a texture generation pipeline, while the influence of geometric control strength at inference time on shape and texture quality remains underexplored.

2.2. Adapter-augmented Diffusion Models and Geometric Control

Adapter modules have become an important mechanism for introducing additional conditions into diffusion models. T2I-Adapter and ControlNet inject spatial conditions such as edges, depth maps, and normals into a pretrained diffusion model through lightweight parallel branches with a controllable scaling factor [24, 25]. IP-Adapter introduces image prompts through decoupled attention [26]. LoRA and its variants implement lightweight model adaptation through low-rank residual updates [27, 28, 29]. Although these methods differ in architecture, they generally include some form of scaling factor that controls the influence of the external condition or adapter residual on the base model.

In practice, the adapter scale is often treated as an empirical hyperparameter. A larger scale usually improves condition adherence, for example by better following normals, depth

maps, or reference images. In texture generation tasks, however, overly strong conditional constraints may weaken the base diffusion model's ability to synthesize color, material, and high-frequency details. Adapter scaling is therefore not a simple matter of increasing the strength as much as possible; it involves a balance between geometric constraints and texture fidelity. This paper focuses on exactly this inference-time control problem: given a fixed adapter structure and fixed geometric conditions, how should the adapter strength be set or scheduled to preserve both shape consistency and texture detail?

2.3. Diffusion Guidance Scheduling and Timestep Control

The denoising process of diffusion models is highly stage-dependent. Early steps mainly determine global layout and coarse structure, middle steps refine shape and semantic structure, and late steps are more responsible for color, texture, and high-frequency details [30, 31]. Based on this observation, many guidance scheduling strategies have been explored in both research and practice. For example, classifier-free guidance scales may be adjusted linearly, with cosine schedules, or dynamically to mitigate oversaturation, detail loss, or reduced diversity caused by overly strong guidance [32, 33]. In addition, the development of latent diffusion, high-resolution diffusion, and Transformer-based diffusion models further shows the close relationship among generation quality, noise schedules, conditional injection, and network architecture [34, 35, 36, 37, 38, 39, 40, 41].

These methods all exploit the staged nature of denoising to regulate conditional strength. Notably, Kynkäänniemi et al. show that classifier-free guidance is harmful at high noise levels, largely unnecessary at low noise levels, and beneficial mainly in a middle interval, motivating limited-interval guidance [14]; mid-stage concentration of conditioning strength therefore has a close precedent in CFG scheduling. However, different control signals act through different mechanisms. CFG scheduling adjusts the weight between conditional and unconditional predictions, mainly affecting the balance among semantic control, diversity, and visual fidelity. By contrast, the scale of a geometric adapter acts directly on intermediate features or residual branches, controlling how strongly geometric conditions intervene in the generation process. TCAS differs from limited-interval guidance in both mechanism and target: it scales adapter feature residuals rather than score extrapolation, raises the middle scale above a conservative baseline instead of switching guidance on and off, and addresses the shape–texture trade-off of geometry-conditioned texture generation, where late-stage over-control suppresses the synthesis of color and high-frequency texture. ControlNet-style scale control is closer in spirit, but it is still usually interpreted as a condition-adherence strength; directly applying a larger, smaller, or smoothly decayed control scale does not specify which denoising stage should receive strong geometric residuals for texture generation.

This distinction matters because the desired schedule for geometry-controlled texture generation is not necessarily monotonic. A monotonic decay or warm-up can reduce over-control in one part of the trajectory, but it cannot simultaneously avoid unstable early residuals, strengthen the middle stage where geometry refinement is most useful, and weaken late residuals that compete with color and high-frequency texture formation. Thus, timestep-conditioned scheduling of adapter strength cannot be treated as a direct transfer of CFG scheduling or generic ControlNet scale scheduling; it requires a dedicated analysis of how geometric residual injection interacts with texture synthesis.

3. Method

3.1. Task Definition and Base Pipeline

This paper builds on the MVPainter-style geometry-controlled multi-view texture generation framework [13] and studies how the inference-time strength of geometric condition injection affects the generated results. Figure 1 illustrates the overall pipeline: the adapter injects geometric residuals into UNet features, TCAS adjusts their strength across denoising stages, and the final multi-view outputs are projected and baked into a textured 3D mesh. Given a single reference image and the corresponding 3D geometry, the base pipeline first generates RGB texture images from six target views and then obtains a textured mesh through projection or texture baking. We do not modify the base UNet, VAE, or pretrained adapter weights. Instead, the adapter scaling factor is treated as the key inference-time control variable.

During multi-view diffusion, the 3D geometry is rendered into normal maps, depth maps, and foreground masks, denoted as:

$$\mathbf{G} = \{G_{\text{normal}}, G_{\text{depth}}, G_{\text{mask}}\}. \tag{1}$$

The geometric adapter maps these conditions to a residual correction term and injects it into the intermediate hidden states of the UNet. For the t-th denoising step, the adapter injection can be written as:

$$\mathbf{h}'_t = \mathbf{h}_t + s(p_t) \cdot A_\phi(\mathbf{h}_t, \mathbf{G}), \tag{2}$$

where $\mathbf{h}_t$ and $\mathbf{h}'_t$ denote the hidden states before and after injection, p_t denotes the sampling progress, and $s(p_t)$ is the adapter scaling strength at the current denoising stage. This formulation highlights a key difference from CFG-style guidance: changing $s(p_t)$ does not reweight two predicted scores, but changes the magnitude of a geometry-conditioned residual added to the feature stream. A larger s usually enhances geometric condition adherence. However, if strong residuals are continuously applied in the late denoising stage, they may suppress the synthesis of color, material, and high-frequency details by the base diffusion model. This paper therefore studies how to schedule s during denoising to balance shape consistency and texture detail preservation.

3.2. Shape-Texture Diagnostic Metrics

The influence of adapter scaling on generated results cannot be evaluated by a single structural metric. We divide evaluation metrics into three groups. The first group consists of shape and structure metrics, including FG-SSIM, Edge-SSIM, and

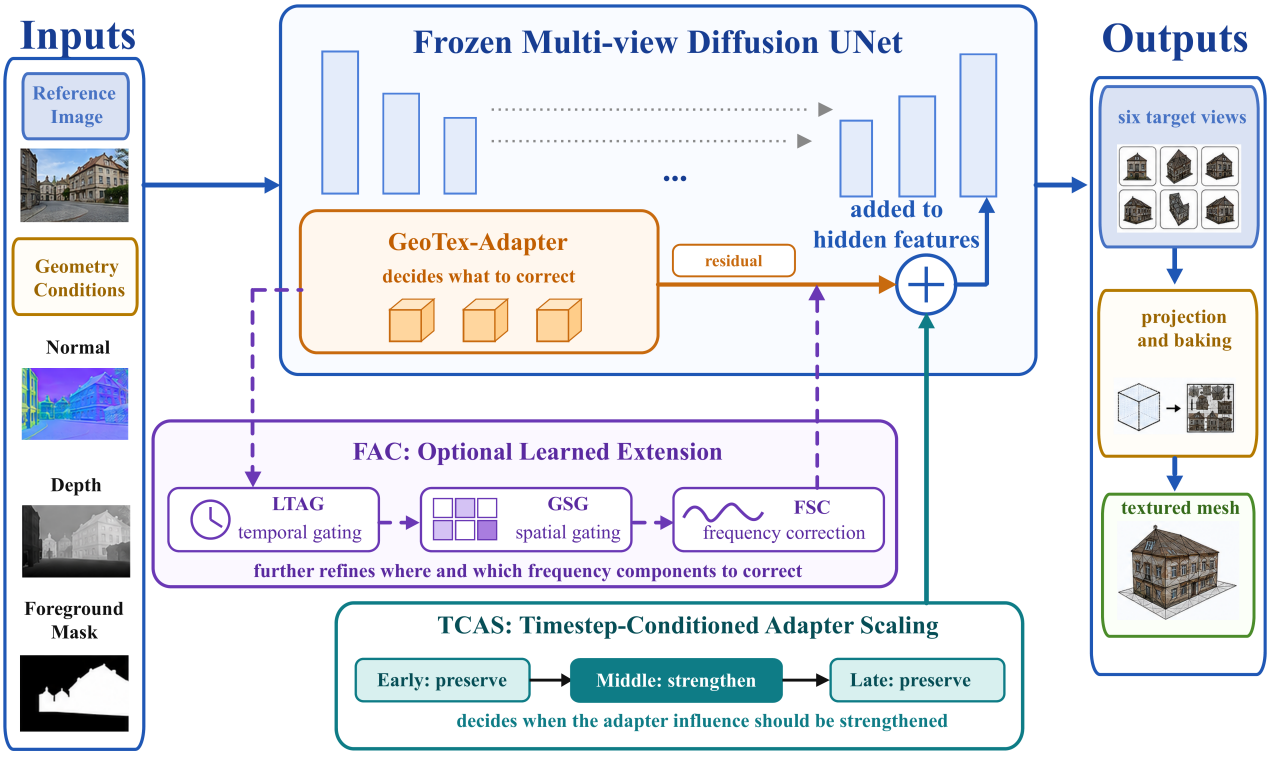


Fig. 1. Method overview. GeoTex-Adapter generates geometric residuals and injects them into UNet features. TCAS dynamically adjusts the residual strength according to denoising stages to balance geometric consistency and texture fidelity. FAC is an optional learned extension module. The final multi-view outputs are projected and baked into a textured 3D mesh.

Crop-SSIM, which measure consistency with ground-truth renderings in the foreground, edge regions, and local foreground crops based on the structural similarity index [42]. The second group contains signal fidelity and perceptual similarity metrics, including PSNR and the learned perceptual image patch similarity (LPIPS) [43], which reflect pixel-level error and differences in deep feature space. The third group consists of texture diagnostic metrics, including Laplacian variance, RGB standard deviation, and gradient magnitude, which are used to detect weakened color variation, surface smoothing, and high-frequency detail loss.

Taking Laplacian variance as an example, the foreground texture response is defined as:

$$M_{\text{tex}} = \text{Var}(\text{Lap}(I_{\text{out}} \odot G_{\text{mask}})), \tag{3}$$

where Lap denotes the Laplacian filter, G_{mask} denotes the foreground mask, and Var denotes spatial variance. A larger M_{tex} usually corresponds to richer local high-frequency variation. If structural metrics improve while Laplacian variance, RGB standard deviation, or gradient magnitude decreases, the improvement may mainly come from contour smoothing or boundary convergence rather than true texture quality improvement.

3.3. Timestep-Conditioned Adapter Scaling

The diffusion denoising process is stage-dependent: early steps mainly establish the global layout and coarse contours; middle steps refine geometric structure and cross-view alignment; late steps are more responsible for color, material, and local texture synthesis. Based on this observation, we propose Timestep-Conditioned Adapter Scaling (TCAS), which replaces a globally fixed adapter scale with a stage-dependent scale function. TCAS is not intended as a generic smooth guidance schedule. Its low-high-low form encodes the adapter-specific hypothesis that geometry residuals should be strongest only after coarse structure becomes stable and before late texture synthesis dominates.

To avoid ambiguity caused by different raw timestep directions in different schedulers, we use sampling progress p to represent the denoising process, where $p = 0$ denotes the start from noise and $p = 1$ denotes the end of sampling close to the final image state. TCAS divides denoising into early, middle, and late stages:

$$s(p) = \begin{cases} s_e, & 0 \le p < 1/3 \\ s_m, & 1/3 \le p < 2/3 \\ s_l, & 2/3 \le p \le 1 \end{cases}. \tag{4}$$

The selected C3 configuration used in this paper is:

$$(s_e, s_m, s_l) = (1.25, 2.50, 1.25). \tag{5}$$

This design follows three principles. First, the early stage uses a conservative scale to avoid disrupting global structure formation with strong geometric residuals. Second, the middle

stage uses a higher scale to strengthen geometric constraints from normals, depth maps, and foreground contours. Third, the late stage reduces the scale again to prevent geometric residuals from suppressing color variation, material detail, and high-frequency texture synthesis. This non-monotonic schedule is therefore different from simply decaying or increasing the adapter strength across all timesteps. TCAS introduces no new parameters and does not modify the base model or adapter weights; it only replaces the scaling factor during inference, so its computational overhead is negligible.

How the stage schedule is selected: the CAI interpretation. We use Correction–Alignment–Interference (CAI) as an empirical diagnostic, not as a theorem or a guaranteed selector. For the main adapter, a 24-object probe was used to nominate C3 before the strict-276 evaluation; the later stage-placement run is explicitly a targeted follow-up. The stage utility notation $U_k = Q(\text{high scale only in } k) - Q(L)$ is useful for describing these results, but additive stage effects are only an approximation because denoising stages interact. Texture diagnostics are treated as variation indicators; fidelity is assessed separately with masked perceptual and color-error metrics. In the independent MV-Adapter audit, the pre-specified CAI rule is set-valued/undefined, so no unique schedule is claimed.

Remark 1. *The CAI interpretation is adapter-dependent. Changing the adapter, residual magnitude, scale semantics, or backbone requires re-estimating the stage utilities and repeating the probe selection. Confidence intervals describe uncertainty in the measured utilities; they do not establish equivalence or non-inferiority. The independent-backbone experiment therefore tests whether stage-position effects are observable under another implementation, not whether C3 or LHL is a universal choice.*

3.4. GeoTex-Adapter and FAC Adaptive Extension

GeoTex-Adapter, TCAS, and FAC correspond to three complementary dimensions of adapter control. GeoTex-Adapter determines what geometric residual correction should be injected. TCAS determines when the residual should be strengthened or weakened along the denoising-time dimension. FAC further modulates the residual in a fine-grained manner along spatial positions and frequency components.

GeoTex-Adapter is used as a fixed geometric residual backbone. It receives a 5-channel geometric input consisting of a 3-channel normal map, a 1-channel depth map, and a 1-channel foreground mask. A multi-scale geometric encoder produces a feature pyramid, and residual corrections are injected into multiple resolution levels of the UNet. TCAS acts on the global temporal scale of these residuals to control the influence of geometric conditions at different denoising stages without changing the adapter architecture.

To further examine whether stage-wise adapter control can be extended to learned adaptive correction, we also conduct experiments with Full Adaptive Correction (FAC). FAC contains three lightweight modules: LTAG learns timestep-dependent gating, GSG generates spatial gates from geometric features, and FSC performs frequency-selective correction on adapter residuals. Unlike fixed TCAS, FAC finely modulates adapter residuals in the temporal, spatial, and frequency domains. In the extension experiments, FAC uses TCAS as the base temporal schedule and further adds GSG and FSC for spatial and frequency correction.

It should be emphasized that FAC requires learning a small number of additional parameters. Therefore, TCAS, which requires no training, remains the main method in this paper. FAC is used only as a learned adaptive extension; Section 4.4 reports a controlled re-examination that delimits when learned modulation can be expected to help.

4. Experiments

4.1. Experimental Setup

4.1.1. Experimental protocol and comparisons

Experiments are conducted using an MVPainter-style geometry-controlled multi-view diffusion pipeline. Each object contains a single reference image and corresponding 3D geometry. During preprocessing, the 3D geometry is rendered into target-view normal maps, depth maps, foreground masks, and ground-truth RGB images. The corrected round-two protocol uses six unique target views, $(0, 15, 12, 16, 13, 14)$ in the forward order (and $(14, 15, 0, 16, 12, 13)$ after reversal); the legacy order with a duplicated top view is retained only as an audit condition. All comparisons with ground-truth renderings use the same views, 256×256 resolution, and foreground masks.

Except for the FAC extension experiment, the saved clean-v2 scaling records keep the object list, fetched reference/conditioning batch, target views, PyTorch initial latent, base UNet, VAE, adapter weights, and number of denoising steps fixed within each object; only the adapter scale or schedule is changed. Those historical records predate explicit Python/NumPy dataset-side seed control, so they are not presented as exact-repeat determinism evidence. All new reruns use Python, NumPy, and PyTorch seeds fixed before dataset construction and before each object. The number of sampling steps is 50. The historical 24-object probe nominated C3 before the larger evaluation records were generated; the active round-two claims use the strict-276 clean-v2 panel and the separately labeled 276-object stage-placement follow-up. The probe-to-holdout protocol isolates the effect of adapter scaling from changes in model architecture, training data, sampling budget, or evaluation-set tuning.

Accordingly, the comparisons in this paper are comparisons among inference-time adapter scaling strategies under the same base pipeline, rather than a system-level benchmark among different texture generation architectures. Uniform adapter scales serve as the primary deployment baselines because they correspond to the most direct way of choosing a single fixed adapter strength at inference time. The probe variants further test whether the trade-off can be resolved by simple alternatives such as globally high scales, layer-wise redistribution, or moving high adapter strength to early, late, or multiple denoising stages. TCAS changes only how this strength is scheduled across denoising stages.

4.1.2. Evaluation sets and metrics

The source protocol records four evaluation sets: (1) a 50-object scale-sweep set; (2) a 26-object texture-audit set; (3) a 24-object probe set for the original schedule nomination; and (4) a 300-object evaluation pool containing a 276-object held-out subset. These exploratory records are retained for provenance; the active round-two results are the strict-276 clean-v2 panel, the separately labeled stage-placement follow-up, and the explicitly bounded extension audits.

Evaluation metrics are divided into three categories: shape and structure metrics (FG-SSIM, Edge-SSIM, and Crop-SSIM), signal and perceptual metrics (PSNR, masked LPIPS, and, where available, CIEDE2000), and texture-variation diagnostics (Laplacian variance, RGB standard deviation, and gradient magnitude). The three variation diagnostics are not direct measures of texture fidelity; their symmetric log-ratio errors relative to ground truth are reported separately. All image metrics are first computed over six unique target views and then averaged over views and objects. For texture-related conclusions, we report means, paired object-level bootstrap intervals, and win rates; DISTS is unavailable in the current baking environment and is not claimed.

The FAC extension uses the same object list and evaluation protocol as the 300-object evaluation pool, and its lightweight modules are trained only on the 1,706-object training pool for the strong-residual adapter instance, which is disjoint from all evaluation objects. TCAS is used as the main baseline, while LTAG, LTAG+GSG, and Full FAC are evaluated as ablation variants under the same paired protocol. The base model and original adapter remain frozen, and only lightweight adaptive correction parameters are learned.

4.1.3. Implementation and dataset details

All objects are drawn from the Objaverse 3D asset collection: ground-truth multi-view images are rendered offline with Blender (Cycles engine, 17 views at 256×256, white background), and the corrected target-view protocol uses the six unique camera IDs $(0, 15, 12, 16, 13, 14)$; the legacy duplicate-top order is retained only as an audit condition. The main adapter's training pool contains 1,118 rendered objects, while the separately documented strong-residual/FAC audit uses a 1,706-object pool from the same source. Both training pools are disjoint from the evaluation pool; we verified that each training/evaluation identifier intersection is zero. Within the 300-object evaluation pool, the 24-object probe set corresponds to the first 24 identifiers (obj_{0000}–obj_{0023}) and the remaining 276 identifiers (obj_{0024}–obj_{0299}) form a holdout that is strictly disjoint from the probe. The exploratory 50-object and 26-object records are retained as provenance; the active evidence freeze uses the strict 276-object cohort and its paired follow-up.

The base pipeline is an MVPainter-style geometry-controlled multi-view diffusion model built on a joint six-view latent UNet, loaded from the local `MVPainter_Pipeline` snapshot under `checkpoints/hf_repo` with diffusers 0.20.0, and used frozen together with its VAE. The pipeline is reference-image-conditioned; no text prompts are used. The GeoTex-Adapter takes a five-channel geometric input (3-channel normal map, 1-channel depth map, 1-channel foreground mask), produces a multi-scale residual pyramid, and injects corrections into multiple UNet resolution levels; for the adapter studied in the main tables, the requested scale is applied multiplicatively at every injected layer. Inference uses the Euler discrete scheduler with 50 denoising steps and a fixed random seed (42) shared by all schedules within an experiment, so all variants share the same initial latents; the model generates six views per object.

Table 1. Strict-276 main-adapter clean-v2 panel. Metrics are pre-save float-evaluation means over the same six unique views; the Full-SSIM column is retained as an audit value and is not silently substituted for the saved-artifact branch.

Method	Full-PSNR↑	Full-SSIM↑	Full-LPIPS↓	FG-PSNR↑	FG-SSIM↑	FG-LPIPS↓	Edge-SSIM↑
No adapter	10.514	0.726	0.627	8.776	0.478	0.207	0.479
Fixed-low	15.086	0.850	0.194	7.030	0.355	0.202	0.500
Fixed-high	13.365	0.823	0.190	5.410	0.229	0.210	0.494
C3 (LHL)	14.875	0.855	0.190	6.763	0.348	0.201	0.500

FG-SSIM, Edge-SSIM (edge masks derived from depth/normal discontinuities), PSNR, and LPIPS (AlexNet backbone) are computed on foreground-masked regions after background normalization; the three foreground variation diagnostics are computed on foreground pixels and are not interpreted as texture fidelity without their GT-relative errors. The main tables use GeoTex-Adapter `geotex_refattn_v1/geotex_step_0002000.pt` trained on the 1,118-object pool. The separate strong-residual/FAC records are labeled as an independent extension protocol and are not pooled with the clean-v2 main-adapter tables. Checkpoint and object-list hashes are recorded in the round-two evidence freeze and reproducibility package.

Statistical reporting. Object-level paired comparisons report 95% percentile bootstrap confidence intervals with 10,000 resamples, a fixed seed, and object win rates; the direction is reversed for lower-is-better metrics. A confidence interval crossing zero is described as no detected difference, not as evidence of equivalence or non-inferiority. Where a mean is accompanied by ±, the quantity is the standard deviation across objects. The stage-placement follow-up is labeled as such and is not treated as a blind confirmation.

4.2. Strict holdout and stage-placement evidence

The current main-adapter evidence is reported in two explicitly separated panels. The first is the strict-276 slice of the clean-v2 four-condition run (no adapter, fixed-low, fixed-high, and C3), evaluated with the pre-save float metric path. The second is a targeted stage-placement follow-up on the same 276 object IDs, with fixed-mean, HLL, LHL, and LLH generated in one run. The repeated C3 outputs are not identical across these two runners (for example, the mean absolute Full-SSIM difference on the common objects is 0.00395), so absolute values are not pooled across panels. All paired comparisons within a panel use the same object IDs and seed.

The clean-v2 panel is a non-dominance result: C3 is above fixed-high on foreground PSNR/SSIM and is close to fixed-low on foreground LPIPS and Edge-SSIM, but fixed-low has higher Full-PSNR, FG-PSNR, and FG-SSIM. Thus the main run does

Table 2. Stage-placement follow-up on the strict 276-object cohort. These are the six non-SSIM metrics from one paired run; LHL, HLL, and LLH use 17/16/17 early/middle/late steps. Lower is better for LPIPS.

Schedule	Full-PSNR↑	Full-LPIPS↓	FG-PSNR↑	FG-SSIM↑	FG-LPIPS↓	Edge-SSIM↑
Fixed mean	14.311	0.188	6.257	0.316	0.205	0.499
HLL	13.397	0.202	5.583	0.259	0.212	0.464
LHL (C3)	14.868	0.190	6.760	0.348	0.201	0.500
LLH	15.136	0.181	6.937	0.357	0.200	0.536

Table 3. Saved-artifact Full-SSIM in the stage-placement follow-up. Prediction is PNG-reloaded float32; GT is the original RGBA composited over white in float32. C3 deltas use object-level paired bootstrap with 10,000 resamples and seed 20260928.

Schedule	Full-SSIM↑	C3 paired difference
Fixed mean	0.87395	+0.00749
HLL	0.85993	+0.02151
LHL (C3)	0.88144	reference
LLH	0.88313	−0.00170

not support a blanket claim that TCAS improves practical image quality over the conservative fixed-low baseline.

The follow-up provides evidence that stage placement matters, but it does not establish LHL as a universal choice. LLH is better than LHL on all six metrics in Table 2 under their intended directions. This is the central counterexample to interpreting CAI as a unique LHL selector. The nominal budgets are also not strictly equal: fixed mean has scale mean 1.6667 and squared-scale sum 138.8889; HLL and LLH have means 1.675 and squared-scale sum 157.8125; LHL has mean 1.650 and squared-scale sum 153.125. The recorded actual residual L2/RMS summaries show further stage-dependent differences, so the result is a stage-placement sensitivity study rather than a perfectly equal-effective-budget intervention.

The corresponding 95% CIs for C3 minus fixed mean, HLL, and LLH are $[+0.00674, +0.00826]$, $[+0.01984, +0.02319]$, and $[-0.00205, -0.00134]$, respectively. The last interval is a detected loss for C3 on this saved-artifact Full-SSIM branch, not evidence that the schedules are equivalent. A 12-object saved-tensor decomposition fixes the GT branch before comparing representations: A fp16 to A fp32 changes mean SSIM by +0.03198; prediction PNG quantization changes it by −0.00036; and GT PNG serialization changes it by −0.00007. These diagnostic effects are not used to retrofit the historical 300-object table.

4.3. Baking and unseen-view case study

To test whether image-level differences survive into a real 3D output, we performed a CPU bake on the frozen, stratified 12-object Exact-GLB cohort. Each of the four main-adapter conditions produced a textured GLB, yielding 48 object–condition exports and 11 unseen-camera renders per export (528 view-level rows). The six source views were fixed and the unseen-view metrics were aggregated within object before any descriptive comparison.

This case study does not show a C3 advantage over fixed-low or no adapter for unseen-view fidelity: no adapter has the best descriptive PSNR, CIEDE2000, and LPIPS means in this cohort. The bake audit verifies semantic mesh/polygon/UV preservation and embedded textures, but it is not a byte-identity guarantee; Blender reindexed vertices for `obj_0048`. The zero seam values are reported with the audit's seam definition and denominator and are not treated as proof of universal seam-free output. Coverage and post-inpainting coverage are reported separately in the supplementary audit. The run has no generated 12-object GT bake and no DISTS result, and visual inspection found cross-view/source-fusion colour inconsistency. Thus the 3D result is a feasibility and limitation case study, not evidence of population-level TCAS superiority. Supplementary S4 gives object-level failure boundaries: `obj_0078` and `obj_0082` have raw coverage below 0.04, while `obj_0048` has 0.8441 inpainted fraction and is the exporter reindexing case.

Table 4. Descriptive unseen-view baking metrics for the 12-object stratified case study. These values are not a population estimate; DISTS was unavailable.

Method	Masked PSNR↑	CIEDE2000↓	FG-LPIPS↓	Silhouette IoU
No adapter	11.824	18.200	0.156	0.99968
Fixed-low	6.001	48.623	0.199	0.99968
Fixed-high	3.891	60.420	0.216	0.99968
C3 (LHL)	5.638	51.183	0.201	0.99968

4.4. FAC Adaptive Correction Extension: A Controlled Re-examination

TCAS uses a fixed three-stage scale, which has the advantages of requiring no training, being simple to implement, and introducing low computational overhead. However, its residual scaling is still uniform across spatial positions and frequency components, lacking fine-grained adaptivity. To examine whether stage-wise adapter control can be extended to learned adaptive correction, we trained the FAC extension—LTAG for timestep gating, GSG for spatial gating, and FSC for frequency-selective correction on top of the TCAS temporal schedule. The base model, VAE, and GeoTex-Adapter remain frozen; only the three lightweight controllers are trainable, adding roughly 6×10^3 parameters. The controllers are trained with the same enhanced denoising objective as the base adapter—a latent noise-prediction MSE with 1.7× foreground and 1.3× edge spatial reweighting plus a latent-SSIM term—using AdamW (peak learning rate 10^{-4}, 100 warm-up steps) for 2,000 steps on the adapter's training pool, which is strictly disjoint from all evaluation objects, and all variants are evaluated under the separately documented strong-residual, per-layer-capped audit protocol with identical objects, seeds, and evaluation code.

In this controlled setting, no learned variant reached the training-free schedule. We tested seven configurations spanning joint, envelope-preserving, fidelity-weighted, and gently scheduled controllers. Their paired deficits relative to the TCAS baseline range from −1.1 to −2.9 dB foreground PSNR (FG-PSNR; win rates at most 16%). A warm-start control in which the gate is left untrained at the exact C3 envelope shows no detected difference from TCAS (paired FG-PSNR $\Delta = -0.18$ dB on the 12-object control, 95% CI $[-0.42, +0.05]$, crossing zero); this interval does not prove equivalence or validate transfer. No configuration or hyperparameter was selected

on any evaluation object; the complete dose–response study is provided in the supplementary material.

For transparency, we note that an initial evaluation of the FAC extension, conducted on the main adapter under our earlier protocol before the strictly paired and strictly disjoint training controls were adopted, had suggested a positive gain. That initial comparison differed in both evaluation protocol and adapter instance, and it does not meet the controls we now consider necessary for a transfer claim; under the strictly paired, strictly disjoint protocol above, no trained configuration replicates the gain. We therefore report only the controlled re-examination and treat the initial comparison as superseded.

The outcome delimits the tested extension and does not identify a causal mechanism: every trained configuration degraded the measured fidelity under this protocol, while the untrained envelope control did not show a detected difference. Learning a modulation that improves upon C3 therefore requires a training signal that protects foreground fidelity; this is future work. The FAC results are reported as a negative, separate extension study, not as a claim that TCAS is the only viable method.

4.5. Second-backbone audit and contribution boundary

We audited the same stage-position question on an independent MV-Adapter implementation using an Exact-GLB cohort of 76 objects and 11 recorded conditions. The full table and paired bootstrap output are provided in the reproducibility package. Dynamic schedules with high scale 1.50 and 1.00 are kept as separate scale families; their rows are not pooled. The table below shows the paired schedule comparison and the conservative controls.

The audit supports an observable stage-position effect within this backbone, but not a uniform practical gain. LHL has slightly higher FG-SSIM and Edge-SSIM than HLL and LLH in the paired schedule comparison, while LLH has the highest PSNR among those four and lower GT-relative texture error than LHL. LHL has no clear advantage over fixed-low or fixed mean under the paired confidence intervals. The frozen CAI rule is `undefined_set_valued`, and the official MV-Adapter pretraining UID list was unavailable, so pretraining disjointness is not asserted. We therefore treat this result as a within- backbone diagnostic, not as a cross-backbone causal validation or a unique schedule-selection result.

5. Limitations

This paper mainly focuses on inference-time adapter scaling and does not retrain the base multi-view diffusion model or the adapter. The evidence supports a bounded stage-placement effect within the tested protocols, not a universal optimizer. The main clean-v2 run does not dominate fixed-low, and the stage follow-up favors LLH over LHL on all six non-SSIM metrics and on saved-artifact Full-SSIM.

Second, Laplacian variance, RGB standard deviation, and gradient magnitude are proxy metrics for texture degradation. They are effective for detecting high-frequency loss and surface smoothing, but they cannot fully replace human perceptual evaluation. For objects with strong reflection, transparency, thin structures, or highly stylized textures, discrepancies may exist between these metrics and subjective quality.

Table 5. Selected within-backbone MV-Adapter means on the Exact 76-object cohort. Absolute scores are not compared with the MVPainter-style backbone.

Method	PSNR↑	FG-SSIM↑	Edge-SSIM↑	FG-LPIPS↓	ΔE_{00} ↓	GT-texture↓
No geometry	13.114	0.498	0.233	0.183	20.720	1.701
Fixed-low	13.357	0.528	0.240	0.188	20.150	2.224
Fixed mean	13.337	0.528	0.240	0.188	20.197	2.203
HLL	13.321	0.527	0.240	0.188	20.243	2.232
LHL	13.344	0.528	0.240	0.188	20.182	2.269
LLH	13.353	0.528	0.240	0.188	20.157	2.199

In addition, the current TCAS uses fixed stage boundaries and a fixed scale combination, C3 = (1.25, 2.50, 1.25). The actual 50-step partition is 17/16/17, and the nominal scale sums, squared sums, and residual norms differ across schedules; consequently the placement result is not a strict equal-effective-budget causal test. The second-backbone CAI rule is undefined/set-valued, and its official pretraining UID disjointness is unknown. The schedule has not been validated across seeds, sampler families, object complexity, material type, or reference-image quality. Future work could study budget-matched stage controls, content-adaptive boundaries, and pre-registered scale-selection rules.

Finally, our controlled FAC re-examination is a separate strong-residual extension protocol. It is negative under the paired controls and is not used to claim that TCAS is universally superior to learned modulation.

6. Conclusion

This paper presents a controlled analysis of adapter scaling in multi-view diffusion texture generation. The evidence shows that stage placement changes both structural and fidelity metrics, and that the effects are adapter- and protocol-dependent. Stronger control can improve selected structural targets while degrading other fidelity or texture measures.

We formulate Timestep-Conditioned Adapter Scaling (TCAS) as a simple low–high–low inference intervention. The 24-object probe nominated C3 for the main adapter, while the strict-276 clean-v2 and stage-placement panels provide the current evidence boundary. Within the follow-up, LHL improves over fixed mean and HLL on several targets, but LLH is better on all six non-SSIM metrics and on saved-artifact Full-SSIM.

Therefore, the defensible contribution is a reproducible, training-free stage-utility/mechanism study: it exposes a shape–texture trade-off, supplies a concrete residual-scaling implementation, and documents both positive stage effects and counterexamples. It does not establish a unique CAI selector, a universal LHL optimum, cross-backbone causality, or a population-level 3D quality advantage. The results motivate treating adapter strength as a stage-dependent control variable whose useful schedule must be re-estimated for each adapter and protocol.

The 12-object baking study confirms that the full GLB and unseen-view path is operational, but its descriptive metrics favor no adapter over C3 and fixed-low for this stratified cohort.

This negative practical result is retained as part of the paper's scope rather than hidden as a selection effect.

Data availability

All data included in this study are available upon request by contact with the corresponding author.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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